Teamwork V (Project Requirements Specification)

Team Name: B

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1. Identify the Stakeholders and User Needs

The main stakeholders of our Hotel Management System are:

Stakeholder	Needs and responsibilities
Receptionist	Manage guests, create bookings, check room availability, and handle check-ins and check-outs.
Hotel manager	View hotel occupancy, room status, bookings, and business summaries.
Administrator	Manage user accounts, permissions, room information, and system settings.
Guests	Provide their information, make reservations, and receive booking details.
Hotel staff	Update room status, such as cleaning and maintenance.

2. Define the Overall Application Requirements

2.1 Functional requirements

1. The system must display a dashboard with hotel statistics.
2. The system must allow staff to view and manage hotel rooms.
3. The system must allow staff to create, view, update, and manage bookings.
4. The system must allow staff to register and manage guest information.
5. The system must store payment records and payment statuses.
6. The system must show room availability and status.
7. The system must prevent overlapping bookings for the same room.
8. The system must provide clear messages when an operation succeeds or fails.

2.2 Non-functional requirements

Non-functional requirements describe the quality of the system.

1. The interface should be simple, clear, and easy for hotel staff to learn.
2. Only authorized users should access protected information and functions.
3. The web application should work on modern desktop browsers.

3. Partition the Application into Logical Subsystems

3.1 Dashboard subsystem

Purpose: Show a summary of the hotel's current situation.

Inputs: Room information, bookings, check-ins, and check-outs.

Outputs: Occupancy rate, available room count, today's check-ins and check-outs, and room availability summaries.

Requirements:

- Display important hotel statistics.
- Show room status clearly.
- Display updated information.

3.2 Room management subsystem

Purpose: Manage hotel rooms and their status.

Inputs: Room number, room type, price, and room status.

Outputs: Room lists, room details, and room availability information.

Requirements:

- Add, view, and update room information.
- Support statuses such as available, occupied, cleaning, and maintenance.

3.3 Booking management subsystem

Purpose: Manage guest reservations.

Inputs: Guest, room, check-in date, and check-out date.

Outputs: Booking records, booking status, and reservation details.

Requirements:

- Create, view, and update bookings.
- Associate each booking with a guest and room.
- Check room availability for the requested dates.
- Prevent conflicting bookings.
- Support the hotel's check-in and check-out process.

3.4 Guest management subsystem

Purpose: Store and manage guest information.

Inputs: Guest name, contact details, and other required information.

Outputs: Guest profiles and guest booking history.

Requirements:

- Register new guests.
- View and update guest information.
- Protect personal information from unauthorized access.

3.5 Payment management subsystem

Purpose: Record and manage payments related to bookings.

Inputs: Booking reference, payment amount, date, and payment status.

Outputs: Payment records and payment status information.

Requirements:

- Record payments of bookings.
- Display payment amounts.
- Validate payment information.

4. Quality Function Deployment (QFD)

4.1 Main customer needs

Customer need	Importance (1–5)
Easy to use	5
Fast booking process	5
Correct room availability	5
Secure guest information	5
Clear dashboard information	4
Easy maintenance and updates	3

4.2 Relationship between customer needs and technical requirements

We use the following relationship scores:

5 = Strong relationship

3 = Medium relationship

1 = Weak relationship

0 = No direct relationship

Customer need	Simple interface	Fast API	Booking validation	Database integrity	Automated testing
Easy to use	5	3	1	0	1
Fast booking	3	5	5	3	3
Correct availability	1	3	5	5	5
Secure information	1	1	1	3	3
Easy maintenance	1	3	1	3	5

4.3 Prioritizing the technical requirements

Technical requirement	Rank
Database integrity	1
Booking validation	2
Automated testing	3
Fast API	4
Simple interface	5

5. Draw use-case and sequence diagrams for the application.

